

بإمكانكم الاطلاع على تسجيلات المادة
من خلال منصة جو اكايمي

CHEMISTRY 101

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جو اكايمي

What is the wavelength of the emitted light produced when an electron moves from $n = 3$ to $n = 1$ in hydrogen atom:

$$\frac{1}{\lambda} = 1.097 * 10^7 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$\frac{1}{\lambda} = 1.097 * 10^7 \left(\frac{1}{9} - \frac{1}{1} \right) = -9751111.1$$

$$\frac{1}{\lambda} = 9653600 \dots \dots \dots \lambda = 1.0255 * 10^{-7}$$

Q2: A sample is composed of (38.7 % C), (9.7 % H) and (51.6 % O) by mass, its molar mass is 62.1 g /mole, what is the empirical formula of this sample?

	C	H	O
%	38.7	9.7	51.6
Mass	38.7 g	9.7 g	51.6 g
Mole	$38.7/12 = 3.225$ mole	$9.7 / 1 = 9.7$ mole	$51.6 / 16 = 3.225$ mole
أقل عدد مولات هو (3.225)			
# of atoms	$3.225 / 3.225 = 1$	$9.7 / 3.225 = 3$	$3.225 / 3.255 = 1$
CH₃O			

- Molar Mass of CH₃O = 31 g/mole

$$\text{Ratio of molar mass} = (62.1 / 31) = 2$$

$$\text{Compound} = 2 * (\text{CH}_3\text{O}) = \text{C}_2\text{H}_6\text{O}_2$$

30 ml of 0.12 M HCl solution are required to neutralize 36 ml of NaOH solution, what is the molarity of NaOH solution?

$$M1 = 0.12$$

$$M2 =$$

$$V1 = 30 \text{ ml}$$

$$V2 = 36 \text{ ml}$$

$$M1 * V1 = M2 * V2$$

$$0.12 * 30 = M2 * 36 \text{ ml} \dots \dots \dots M2 = 0.1 \text{ M}$$

Calculate the velocity of moving neutron ($1.67 * 10^{-27} \text{ Kg}$) whose wavelength is 500 pm:

$$\lambda = 500 \text{ pm} = 500 * 10^{-12} \text{ m}$$

$$\lambda = \frac{h}{mv} \implies v = \frac{h}{m\lambda}$$

$$v = \frac{6.626 * 10^{-34} \text{ J} * \text{S}}{1.67 * 10^{-27} \text{ Kg} * 500 * 10^{-12} \text{ m}} * \frac{\text{Kg} * \text{m}^2}{\text{S}^2} = 793.5 \frac{\text{m}}{\text{s}}$$

How many unpaired electrons are there in ground state electronic configurations of phosphorus atom?

$$\text{P (15 electrons)} = 1S^2 2S^2 2P^6 3S^2 3P^3$$

$$\# \text{ of unpaired electrons} = 3$$

Which one of the following elements is metalloid?

Cs

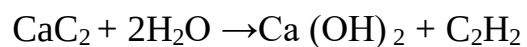
Ge

W

Hg

Answer = Ge

How many grams of H_2O must be consumed in order to produce 6.5 g C_2H_2 according to this equation?



- Molar Mass of $\text{C}_2\text{H}_2 = 26 \text{ g /mole}$
- Mole of $\text{C}_2\text{H}_2 = \text{Mass} / \text{molar Mass} = 6.5 / 26 = 0.25 \text{ mole}$
- $2 \text{ mole H}_2\text{O} \rightarrow 1 \text{ mole C}_2\text{H}_2$
?? Mole $\text{H}_2\text{O} \rightarrow 0.25 \text{ mole C}_2\text{H}_2$
- # of mole $\text{H}_2\text{O} = 0.5 \text{ mole}$
- Mass of $\text{H}_2\text{O} = 0.5 * 18 = 9 \text{ gram}$

The theoretical yield In g of (NH₃) from the reaction of 7.1 g (N₂) and 36 g (H₂) according this equation is (N₂ + 3 H₂→ 2 NH₃)

- Molar Mass of N₂ = 28 g/mole
- Molar Mass of H₂ = 2 g/mole
- Mole of N₂ = 7.1 / 28 = 0.254 mole
- Mole of H₂ = 36 / 2 = 18 mole
- Limiting reactant = N₂
- 1 Mole N₂ → 2 mole NH₃

0.254 mole N₂ → ?? Mole NH₃

Mole of NH₃ = 0.508 mole

Mass of NH₃ = Theoretical yield = 0.508 * 17 = 8.636 g

A certain liquid has density of 2.67 g/cm³, 30.5 L of this liquid would have a mass

$$1 \text{ L} = 1000 \text{ cm}^3$$

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \implies \text{Mass} = \text{Density} * \text{Volume}$$

$$2.67 \frac{\text{g}}{\text{cm}^3} * 30.5 \text{ L} * \frac{1000 \text{ cm}^3}{1 \text{ L}} = 81435 \text{ g} = 81.435 \text{ Kg}$$

How many significant figures are in (0.00340501)?

6 significant figures

750 m/s = Km/hour

$$\frac{750 \text{ m}}{\text{s}} * \frac{3600 \text{ s}}{1 \text{ hour}} * \frac{1 \text{ Km}}{1000 \text{ m}} = 2700 \frac{\text{Km}}{\text{hour}}$$

The energy of 1 mole of photons that has a wavelength of 0.02 m is J

$$E = h * \left(\frac{c}{\lambda} \right) * 6.02 * 10^{23}$$

$$E = 6.626 * 10^{-34} * \left(\frac{3 * 10^8}{0.02} \right) * 6.02 * 10^{23}$$

$$E = 5.983 \text{ J}$$